OMNISPACE – WEEK 4 REPORT

UX Research & Spatial Architecture

Project	OmniSpace – Spatial UI System
Internship	Infotact Solutions – UI/UX Design
Role	UX Research & Spatial Architecture
Team Responsibility	Member 1
Week	Week 4

1. Introduction

OmniSpace is a spatial user interface concept in which digital panels and windows are arranged in a three-dimensional environment.

In a spatial interface, users interact with digital content placed at different distances and positions. Users should be able to locate important panels, understand their purpose and move their attention between them without confusion.

Week 4 focuses on user testing. The purpose is to evaluate the proposed spatial navigation and identify situations where users may lose track of floating windows.

2. Objectives

- Evaluate the usability of the OmniSpace spatial interface.
- Understand whether users can locate floating windows easily.
- Identify navigation difficulties between Near, Middle and Far UI panels.
- Check the visibility and readability of spatial content.
- Identify possible confusion during spatial navigation.
- Recommend improvements based on testing observations.

3. Understanding User Testing

User testing is a method of evaluating an interface by asking people to perform specific tasks.

For OmniSpace, users can be given simple navigation tasks in a spatial interface prototype or a suitable demonstration.

During testing, the researcher observes how users locate panels, understand their positions and move their attention between different interface elements.

The aim is to identify usability problems, not to test the user's ability.

4. Task-Based User Testing Plan

Task 1: Locate a Floating Window

User task: Find a specified floating window in the spatial environment.

What to observe:

- Can the user identify the correct window?
- Does the user search in the wrong direction?
- Does the user need additional guidance?

UX purpose: To understand whether floating windows are easy to locate.

Task 2: Identify the Near UI Panel

User task: Find the Near UI panel containing quick-access controls.

What to observe:

- Can the user identify the closest panel?
- Are the controls visible and understandable?
- Is the panel placed comfortably?

UX purpose: To evaluate the visibility and accessibility of important controls.

Task 3: Locate the Far UI Panel

User task: Find a notification or information panel placed farther away.

What to observe:

- Can the user notice the panel?
- Is the text readable?
- Does the user understand that the panel is farther away?

UX purpose: To evaluate distant information visibility and spatial awareness.

Task 4: Navigate Between Panels

User task: Move attention between Near, Middle and Far UI panels.

What to observe:

- Can the user understand the panel arrangement?
- Does the user become confused when changing focus?
- Are the panels visually distinguishable?

UX purpose: To evaluate navigation between different spatial layers.

Task 5: Track a Repositioned Window

User task: Locate a floating window after its position has changed.

What to observe:

- Can the user find the window again?
- Does the user remember its previous position?
- Is the new position easy to understand?

UX purpose: To evaluate whether users can maintain awareness of moving interface elements.

5. Observation Criteria

Observation area	What to check
Window visibility	Can users notice the floating window?
Navigation	Can users move between panels easily?
Spatial awareness	Do users understand panel positions?
Readability	Can users read text comfortably?
Visual hierarchy	Can users identify important information?
Window tracking	Can users find a repositioned window?

Observation area	What to check
User confusion	Do users hesitate or select the wrong panel?

6. Possible Usability Problems and Solutions

Possible problem	Suggested UX solution
Users lose track of floating windows	Use consistent panel positions and clear labels
Far panels are difficult to notice	Improve contrast and visual visibility
Text is difficult to read	Use suitable text sizes and spacing
Panels look similar	Give each panel a clear visual identity
Too many windows create confusion	Reduce unnecessary panels and visual clutter
Users struggle to understand depth	Maintain clear Near, Middle and Far placement
Repositioned windows are difficult to find	Use clear movement feedback and consistent behaviour

7. Spatial Architecture Evaluation

Spatial architecture describes how digital panels are arranged in a three-dimensional environment. OmniSpace uses three main spatial levels:

7.1 Near UI

The Near UI contains quick-access controls and important actions.

- Important controls should be easy to notice.
- Text and buttons should be readable.
- The panel should be placed within a comfortable interaction area.

7.2 Middle UI

The Middle UI contains the main application content and frequently used information.

- Content should have a clear visual hierarchy.
- The panel should be easy to identify.
- Information should be organized without unnecessary clutter.

7.3 Far UI

The Far UI contains secondary information and notifications.

- Important information should remain visible.
- Text should be readable from the intended viewing position.
- The panel should not distract users from primary tasks.

8. User Testing Flow

- Introduce the OmniSpace interface to the participant.
- Explain the basic navigation task.
- Ask the participant to locate a floating window.
- Ask the participant to navigate between Near, Middle and Far UI panels.
- Ask the participant to locate a repositioned window.

- Observe the participant's actions and difficulties.
- Collect feedback about visibility and navigation.
- Record usability problems and possible improvements.

9. Testing Data Collection

Details	Record
Number of participants	To be filled after testing
Testing date	To be filled
Tasks completed	To be recorded
Navigation difficulties	To be observed
Visibility issues	To be observed
Participant feedback	To be recorded
Suggested improvements	Based on observations

Important: No user testing results are claimed in this report. The table is provided for recording actual findings after testing.

10. UX Recommendations

- Keep important floating windows in predictable positions.
- Use clear labels to identify different panels.
- Maintain sufficient contrast between panels and backgrounds.
- Use readable text sizes at different viewing distances.
- Avoid placing too many panels in the user's field of view.
- Maintain a consistent arrangement for Near, Middle and Far UI.
- Provide clear visual feedback when a window moves.
- Keep secondary information less prominent than important content.
- Evaluate navigation using simple task-based testing.
- Improve the design based on actual user observations.

11. Expected Outcomes

- A task-based user-testing plan.
- A record of navigation and visibility observations.
- Identification of possible floating-window usability problems.
- Recommendations for improving spatial navigation.
- An evaluation of Near, Middle and Far UI arrangement.
- A documented UX research summary for the OmniSpace project.

12. Conclusion

Week 4 focuses on evaluating the OmniSpace spatial interface through task-based user testing. The evaluation helps identify whether users can locate floating windows, understand spatial depth and navigate between different interface panels.

By recording actual observations and improving the design based on user feedback, OmniSpace can provide a clearer, more comfortable and user-friendly spatial interaction experience.

Report scope: Week 4 – Column 1: UX Research & Spatial Architecture. This report covers user testing and spatial navigation evaluation only.